

## Full length article

## Automated text classification of near-misses from safety reports: An improved deep learning approach

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## ABSTRACT

Examining past near-miss reports can provide us with information that can be used to learn about how we can mitigate and control hazards that materialise on construction sites. Yet, the process of analysing near-miss reports can be a time-consuming and labour-intensive process. However, automatic text classification using machine learning and ontology-based approaches can be used to mine reports of this nature. Such approaches tend to suffer from the problem of weak generalisation, which can adversely affect the classification performance. To address this limitation and improve classification accuracy, we develop an improved deep learning-based approach to automatically classify near-miss information contained within safety reports using Bidirectional Transformers for Language Understanding (BERT). Our proposed approach is designed to pre-train deep bi-directional representations by jointly extracting context features in all layers. We validate the effectiveness and feasibility of our approach using a database of near-miss reports derived from actual construction projects that were used to train and test our model. The results demonstrate that our approach can accurately classify 'near misses', and outperform prevailing state-of-the-art automatic text classification approaches. Understanding the nature of near-misses can provide site managers with the ability to identify work-areas and instances where the likelihood of an accident may occur.

## 1. Introduction

Construction is a dangerous industry, which is exacerbated by its temporary nature and innate complexity [36,33,24,38]. In the United States, for example, in 2017 a total of 4674 fatalities were recorded in the workplace, with 971 (20.7%) occurring in construction (20.7%). Moreover, most of these fatalities were attributable to being struck by an object, electrocution, and caught-in/between plant and machinery [30]. Contrastingly in China, a staggering 3843 fatal injuries were recorded in 2017 on construction sites [7]. However, the precursors of these fatalities, accidents and near misses are not being systematically identified during construction [52].

A near miss has been defined as an unplanned event that has the potential to cause but does not result in personal injury, environmental or equipment damage, or interruption to regular operation [32]. Heinrich [15] observed that approximately 91% of accidents produced no injuries, while 9% were minor and less than 1% major. Also, Heinrich

[15] hypothesised that a multitude of near misses is a prerequisite for a workplace injury or fatality. Thus, being able to anticipate the likely precursors of a potential accident would provide site managers with the ability to put in place measures to ensure peoples safety on site.

Safety reports are an extremely valuable source of information that can be used by site managers to learn about the conditions and events that have contributed to the occurrence of accidents and therefore facilitate interventions to ensure positive safety outcomes [13,22]. Within a safety report, near-misses are typically documented in an unstructured or semi-structured free-text data format, which contains information such as the description of the event, its time, and location. The analysis of near-miss data can be labour intensive and time-consuming, and it requires an understanding of safety to be able to derive meaningful insights. With considerable headway being made in the field of artificial intelligence, we can now automatically process, organize and handle free-text data.

Under the auspices of safety management, numerous text

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classification-based approaches been developed to acquire a better understanding of accident causation [5]. However, such studies have tended to focus on developing machine learning-based approaches that consist of: (1) the manual extraction of text features; and (2) inputting these features into a classifier. Goh et al. [13], for example, evaluated six machine learning approaches that were used to classify accidents reports. Goh et al. [13] found that support vector machine (SVM) provided the most accurate and reliable results compared to the use of Linear Regression, Random Forest, k-nearest Neighbor, Decision Tree, and Naïve Bayes approaches. Despite the success of hand-crafted features, they are prone to suffering from the problem of weak generalization, which profoundly affects their classification accuracy [2,23,21]. In contrast to machine learning, ontology-based approaches have been used for text classification as they utilise the semantic features of free-text [49], but are unavailable for specific domains or different languages [1].

Our research aims to improve the accuracy of text classification and enable generalisations about the nature of near-miss reports to be made. Possessing knowledge of the nature of near-misses, which arise during construction can provide site managers with a much-needed understanding of their various guises and frequencies. As a result, this can provide site managers with an ability to identify work-areas and instances where the likelihood of an accident may occur. Against this backdrop we utilise deep learning and Bidirectional Transformers for Language Understanding (BERT) to develop a robust automatic text classification model of near-misses. Essentially, a BERT has a multi-layer bidirectional transformer encoder-decoder structure that possesses a powerful ability to classify texts. The BERT has been demonstrated to not only attain the highest level of accuracy for text classification but is also generalisable across different databases [53].

Our paper commences by reviewing past studies on text classification in construction and examines the developments of deep learning-based text classification in computer science. Then, we develop our deep learning automated text classification model, which is subsequently tested, and its performance evaluated.

## 2. Related works

### 2.1. Text classification in construction

A wide range of digital techniques has been adopted in construction projects to improve collaboration, coordination, and the exchange of information between organizations [8,3,39,14]. Varying data formats are collected, stored and exchanged in construction (e.g., images, video, and text), which are used to support planning, control, and decision making [8]. However, a considerable portion of this data is stored in an unstructured or semi-structured text format [40,8,3,39,14]. It is essential to classify such text to facilitate the management of information and knowledge [51]. Typical approaches that have been used to classify text are presented in Table 1, which can be broadly categorised as being (1) knowledge-based; and (2) machine learning-based.

Ontology has been regarded as one of the most popular knowledge-based approaches. An ontology is a representation of concepts, relationships, and axioms that form a foundation for reasoning about a

domain [18]. There have been numerous ontology-based text classification approaches developed in construction. For example, Chi et al. [6] established an ontology-based text classification approach to support the automation of Job Hazard Analysis (JHA). Similarly, Salama and El-Gohary [41] developed a semantic text classification approach to classify clauses and subclauses for regulatory and contractual compliance in construction.

Advances in machine learning have provided new possibilities for automatic text classification. Several machine learning-based approaches have been investigated for text classification in construction, which have utilised higher-order paths [42,41,1]. Zhang et al. [48], for example, utilised text classification in combination with text mining and natural language process (NLP) to analyse accidents. Likewise, Marucci-Wellman et al. [25] compared four machine learning approaches (i.e., Naïve Bayes, Single word and Bi-gram models, SVM and Logistic Regression) to classify injury narratives contained within large administrative databases.

The accuracy and efficiency of these approaches depend upon the quality of annotated corpora and the features that have been extracted, which enables the effectiveness of the developed corpus-based text classification approach. Furthermore, it is often difficult to adapt a developed approach to solve new problems or in a different language to address the same issue [45]. Put simply, developed machine learning-based text approaches are readily generalisable.

### 2.2. Deep learning-based text classification

Deep learning can extract complex characteristics and learn representations from input data, which have been widely for computer vision applications [10,11,16], as well as text classification [44,47]. An innate feature of deep learning-based text classification is to accurately extract word order information and then further learn from the semantic information contained within the text. Compared with traditional machine learning, deep learning is an end-to-end procedure where representations are learned from an extensive training database without domain expertise for feature extraction. Previous studies have demonstrated that deep learning-based approaches can provide superior accuracy for text classification when compared with traditional machine learning [44].

Deep learning-based text classification approaches require word embedding (i.e. a learned representation for text where words that have the same meaning have a similar representation to transform words into vectors of consecutive numbers). A popular deep learning approach that utilises word embedding is the Word2Vec [28], which is a two-layer neural net that processes text. Its input is a text corpus, and the output is a set of vectors. Word embedding enables the extraction of semantic relations from various textual reports or documents with the assistance of feed-forward artificial neural networks [9,29]. The success of this process, however, is reliant on having access to a significant volume of textual data. To address this issue, external semantic resources (e.g., Wikipedia) are used to label data when distributed approaches' (e.g., Word2vec) are utilised for text classification [1].

To further increase the representation flexibility of deep learning models, attention mechanisms (i.e. they serve to orient perception as

**Table 1**  
Performance comparison of text classification approach.

| Approach                                            | Description                                                                                                                                                                                         | Author (Year)             |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Text mining and natural language processing         | Five text classification models are used to classify causes of accidents text                                                                                                                       | Zhang et al. [48]         |
| Machine learning approach                           | Six machine learning approaches are applied for text classification. The results show that the SVM approach achieved a better performance (e.g., accuracy)                                          | Goh et al. [13]           |
| Machine learning-based text classification approach | A semantic-based machine learning approach is developed to classify clauses and subclauses from textual documents for regulatory and contractual compliance checking in construction                | Salama and El-Gohary [41] |
| Ontology-based text classification approach         | An ontology-based semantic approach is used to improve the text classification performance by utilising the semantic features, which have outperformed the use of a machine learning-based approach | Zhou and El-Gohary [49]   |

well as memory access within a neural network) have been introduced and developed, which form an integral part of the text classification process. The attention module is trained to capture the dependencies that make significant contributions to text classification, regardless of the distance between the elements in the sequence. For example, Yang et al. [47] developed a hierarchical attention network document classification that comprised of two levels that enabled it to attend differentially and import contents when constructing the document representation. Similarly, Shin et al. [43] developed a neural attention mechanism to classify radiology reports, which have outperformed non-neural approaches on an extensive database.

Up until this point in time, the best performing deep learning models have consisted of an encoder and a decoder that are connected through an attention mechanism, with successful applications for sentiment classification [46,12], sentence pair modelling, and summarisation. It should be noted that BERT has been proven to be the most accurate language representation model for text classification when compared with prevailing state-of-the-art approaches and has even surpassed human performance in some tasks [53]. To achieve a higher level of classification accuracy on near-miss reports and enable faster convergence during model training, we develop an improved BERT approach by modifying and adjusting its architecture and hyper-parameters.

### 3. Source of data

Huazhong University of Science and Technology has been collaborating with the Wuhan Metro Group Co., Ltd (China), who are in the process of extending its metro-rail network. Consequently, researchers were provided with safety reports from several construction sites involved with the extending Wuhan's existing metro-rail network. The data were recorded by engineers who have extensive experience in documenting safety incidents, which were categorised in accordance with the Quality and Safety Check Points of Urban Rail Transit Engineering [26], 2011 GB 50715-2011 [27]. The collected data was uploaded and stored in the Wuhan Metro Safety Management System, as is shown in Fig. 1. In this paper, we are only concerned with near-misses that occurred during construction.

In this database system, approximately 3280 near-miss events are stored. Each near-miss contains its location, time, name, description, safety level, categories, and images (Fig. 1). These 3280 near-misses have been classified into 170 categories, such as quality of main concrete structure, template installation, monitoring data overrun. Table 2 presents examples of our created database. The database is randomly divided into a training and testing database with a ratio of 8:2. In other words, 2624 near-miss are used for training the BERT model, and 657 for testing its performance.

## 4. BERT-based text classification approach

### 4.1. Architecture of BERT

The BERT's model architecture is a multi-layer bidirectional transformer encoder-decoder structure [53]. The encoder maps the input sequence of symbol representations  $x = (x_1, \dots, x_n)$  to a sequence of continuous representations  $z = (z_1, \dots, z_n)$ , and then the decoder generates an output sequence  $y = (y_1, \dots, y_n)$  of symbols one element at a time. The structure of the encoder and decoder are presented in Fig. 2. Then, the key elements of the transformer are briefly introduced in the following section. Further details about the structure and nature of the Transformer Model can be found in Vaswani et al. [35] and Devlin et al. [53].

The *encoder* consists of six identical layers. Each layer has two sub-layers: (1) a multi-head self-attention mechanism; and (2) a fully connected feed-forward network with simple and position-wise. These two sub-layers are connected by a residual connection [17] followed by layer normalisation, and then output a 768-dimension vectors. Therefore, the outputs of the sub-layers are denoted as  $o_s$ :

$$o_s = \text{LayerNorm}(x + \text{SubLayer}(x)). \quad (1)$$

The *multi-head attention* sublayer plays an important role in the transformer's architecture. It utilises an attention mechanism, which relates different positions of a single text to it and then computes a better representation, as denoted in Fig. 3. The attention mechanism [31,4] is widely used in Natural Language Processing (NLP), which maps a query vector  $Q$  and a set of key-value vector pairs  $(K, V)$  to an

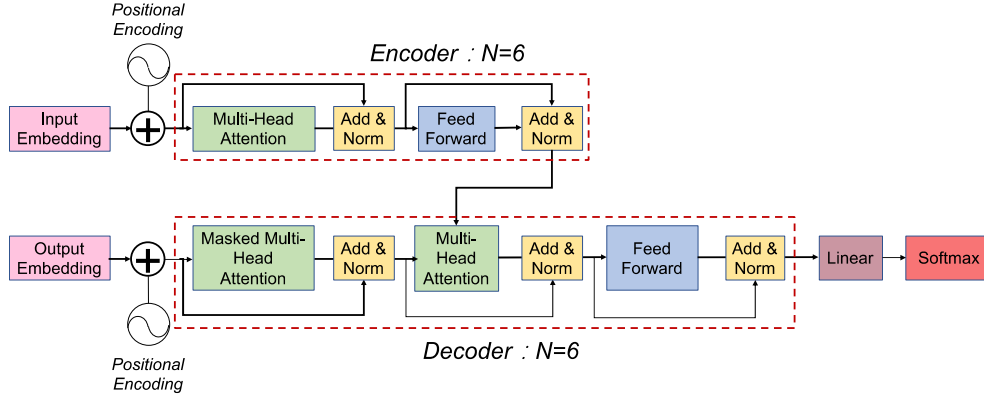
| 隐患排查基本信息                                      |                    |
|-----------------------------------------------|--------------------|
| 编号: NO0002361                                 | Report number      |
| 标段: 027BT002                                  | Site number        |
| 隐患名称: 施工机械碰撞, 损伤支撑结构                          | Near miss category |
| 隐患部位: 纸坊大街北段基坑                                | Near miss location |
| 排查人: 王振华                                      | Reporter           |
| 排查单位: 二十七号线二标                                 |                    |
| Line number: 线路: 27号线                         |                    |
| Site location: 工点: 纸坊大街站                      |                    |
| Near miss level: 隐患等级: 一般隐患                   |                    |
| Near miss description: 纸坊大街北段基坑第一道混凝土支撑上停放挖掘机 |                    |
| Report time: 排查时间: 2016/11/09 11:38:58        |                    |
| 备注:                                           |                    |
| Site pictures of near miss                    |                    |
|                                               |                    |
| Safety requirements for near miss prevention  |                    |
| 整改截止日期: 2016/11/12 23:59:00                   |                    |
| 消除要求: 将纸坊大街北段基坑第一道混凝土支撑上停放的移除挖掘机              |                    |

Fig. 1. Web-based near-miss management system.

Source: [52,p.149].

**Table 2**  
Examples of our created near-miss database.

| Categories                         | Description                                                                                                                                                                  |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Quality of main structure          | The CX7 monitoring point was covered by slag for more than 24 h and was not cleaned up                                                                                       |
| Steel processing and production    | The design of the distribution ribs with a diameter of 18 mm of the bottom ribs of the bottom plate is 150 mm and the actual measurement is 180 mm                           |
| Foundation pit working environment | The bottom suspension of the first ballast support in the 23rd section of the north end of the triangular road station was not removed in time, and there is a safety hazard |
| Frontier protection                | The second warehouse, the north end of the board; there is no protection in place                                                                                            |
| Personal Protective Equipment      | Two workers mixed up on the work (mixed paint) without hanging the safety rope (not back the safety rope)                                                                    |
| Shield tunneling construction      | 151 ring tube of right line has slight leakage                                                                                                                               |
| Pit edge load                      | The distance between the steel support and cofferdam of the foundation pit is less than the design requirement                                                               |



**Fig. 2.** Structure of the Transformer Model [35,p. 6003].

output, described as follows:

$$o_a = \text{Attention}(Q, K, V). \quad (2)$$

The *multi-head attention* employs  $n$  different linear layer to linearly map vector  $Q, K, V$  and then focuses on obtaining multi-head attention outputs as follows:

$$o_m(Q, K, V) = \text{Concat}(h_1, h_2, \dots, h_n)W^o$$

$$h_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V), i = 1, \dots, n \quad (3)$$

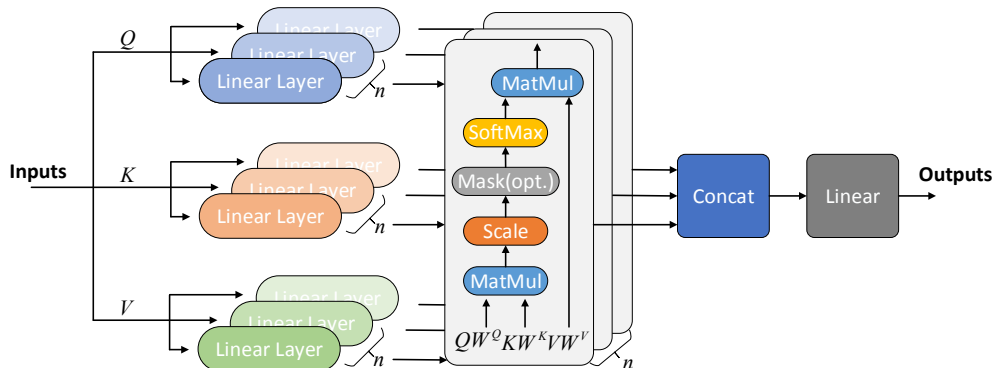
where  $W^o, W^Q, W^K$  and  $W^V$  denotes the linear layer weight for output, vectors  $Q, K$  and  $V$  respectively.

We use the same self-attention mechanism (or intra-attention) and vector  $Q, K, V$  in our work. The self-attention mechanism computes a representation of a single sequence whereby relevant information is utilised by different positions within its structure. Self-attention has been used in a wide range of applications including reading comprehension [4] and abstractive summarization [31]. In our work, all of the

keys  $K$ , values  $V$  and queries  $Q$  are derived from the same place. Self-attention has the following advantages in our task as the:

- total computational complexity per layer is smaller than others such as those that are recurrent and convolutional. The detailed operation is shown in Fig. 3;
- amount of computation that can be parallelised, relative to the Recurrent layer, as measured by the minimum number of sequential operations required; and
- path length between long-range dependencies in the network is short. The CNN needs an additional convolutional layer to expand its perceptual field. Similarly, the RNN needs to compute input from time 1 to  $n$ . However, self-attention only needs one step within the matrix operation.

In this research, the Scaled Dot-Product Attention is used to catch inner-sequence information, which is formulated as:



**Fig. 3.** Multi-head attention with several attention layers in parallel.

Source: [35,p. 6004].

$$o_a = \text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (4)$$

where  $d_k$  denotes the dimension of keys  $K$  and values  $V$ . As in the Eq. (4), the Scaled Dot-Product Attention utilises a Scaled Dot-Product with division  $\sqrt{d_k}$  to calculate similarity, which is the difference between the general attention model for adjustment that prevents the inner product from becoming too large.

The second kind of sub-layer is a fully connected layer that possesses a position-wise mechanism. Since this layer processes the attention position  $i$  in the Multi-head Self-attention, the feed-forward network is referred to as being ‘Position-wise’. The position-wise feed-forward networks have two linear layers with a rectified linear unit (ReLU) layer between them, which is summarised as:

$$o_p = [\max(xW_1^p + b_1^p, 0)]W_2^p + b_2^p \quad (5)$$

In the encoder, the weights  $W_i^Q \in \mathbb{R}^{d_m \times d_k}$ ,  $W_i^K \in \mathbb{R}^{d_m \times d_k}$ ,  $W_i^V \in \mathbb{R}^{d_m \times d_v}$  and  $W^o \in \mathbb{R}^{n d_v \times d_m}$ , where  $d_m = 768$  in this work denotes the dimension of input data. Besides,  $n = 12$ , and  $d_k = d_v = d_m/n = 64$ . We set the feed-forward/filter size to be  $4d_m = 3072$ , according to the network structure in Fig. 2.

The decoder consists of six identical layers that are composed into three sub-layers: (1) masked multi-head attention; (2) multi-head attention; and (3) feed forward network. Similar to the encoder, these three sub-layers are connected by residual connections, which are used to produce a 768-dimension vector. The multi-head attention layer is modified to stack the prevent positions from attending to subsequent ones, which have more inputs than the encoder. The masked multi-head attention ensures that the predictions for position  $i$  are dependent on those less than  $i$  that have temporal causality.

To sum up, the RNN needs to compute in line with the sequence order of inputs with a high computational complexity. However, the RNN is weaker than the transformer in dealing with long-time dependence. In our research, the transformer, an encoder-decoder style model, is adopted instead of an RNN. Notably, the transformer network is unable to obtain temporal information. In this instance, two sequences with the same words, but being ordered differently, will produce the same features. We, therefore, need to embed the position information into the input vector. The final inputs are composed of three parts: (1) token embeddings using *WordPiece* [37], (2) positional embeddings learned from frequencies in Vaswani et al. [35]; and (3) segment embeddings to separate sentence pairs that are packed together. These embeddings are then summed to obtain an input representation.

#### 4.2. Model Pre-training

Model pre-training plays an import role in improving the accuracy and reliability of the developed model. The pre-training phase provides the weight of model with a useful initialisation so that a new task with experience can learn from big datasets. In this research, the pre-training process involves the following two tasks [53]:

- **Task 1: Masked LM:** It is well-acknowledged that a deep bi-directional model is more powerful than the left-to-right model. Thus, to train our bi-directional model, we randomly mask a percentage of the input sentences. Then, we only predict those words that have been masked. Therefore, the model is forced to catch bidirectional contexture information to predict masked words. This results in the training data randomly generating 15% of its tokens as being masked, which follows the setting identified in Devlin et al. [53].
- **Task 2: Next Sentence Prediction:** The relationship between two text sentences is an important feature to understanding the nature of language modeling. The model is asked to predict whether the sentence B is the next sentence of A.

For the pre-training data, we concatenated two corpora: (1) BooksCorpus (800 M words) [50]; and (2) English Wikipedia (2500 M words). With the concatenated corpus, the proposed model is trained with batch size of 256 sequences and 512 tokens in each sequence for 1,000,000 steps. We use the Adam method’s learning rate algorithm  $1e-4$ ,  $\beta_1 = 0.9$ ,  $\beta_2 = 0.999$ ,  $L_2$  weight decay of 0.01, learning rate warmup over the first 10,000 steps, and linear decay of the learning rate. A dropout probability of 0.1 is adopted in each layer [53].

#### 4.3. Modification for classification task

Our research modifies the BERT model for the classification task. We first remove the last Softmax layer in Fig. 2. Since that the number of labels in our dataset is  $d_l = 170$ , an extra linear layer is added after the last layer with weight  $W^l \in \mathbb{R}^{d_m \times d_l}$ . Then, we add the Softmax layer to assign a probability of each label. To fine-tune our model the hyperparameters are the same for pre-training, with the exception of the batch size, learning rate, and number of training epochs. The dropout probability was kept at 0.1 [53]. Detailed hyperparameters are shown in the experiment section.

### 5. Experiments and results

The developed BERT-based text classification approach was performed on a server with an Intel(R) Xeon(R) E5-2670 CPU, NVIDIA(R) GeForce GTX 1080 with 8 GB video memory GPU, and 64 GB RAM. For the purpose of this research the Python programming language was used. The TensorFlow 1.8.0 deep learning framework was drawn upon.

#### 5.1. Data pre-processing

Before training our developed BERT model, the data is pre-processed using the following three steps:

- **Step 1: Data cleaning** - In this step, unrelated text such as punctuation are removed, and each sentence is intercepted by the first  $N$  words. In addition, all words of the sentence are converted to lowercase. Here,  $N$  was set to 64 according to the sentence length in our dataset.
- **Step 2: Wordpiece tokenization** - Wordpiece tokenization is completely data-driven and guaranteed to generate a deterministic segmentation for any possible sequence of characters [37]. In this research, a greedy longest-match-first algorithm is used to perform the tokenization. For example, the input word is “unaffable”. Then output (“un”, “##aff”, “##able”).
- **Step 3: Text-feature construction** - All the different tokens from the previous step are arranged, and then numbered from 1 to  $k$ . If the length of sentence is less than  $N$  ( $k < N$ ), it will be filled with o. In doing so, each sentence is converted to an input feature of length  $N$ . In addition, when the label is processed, the number is used, and 0–( $L-1$ ) represents  $L$  labels. In this research, there were 170 categories ( $L = 170$ ).

#### 5.2. Evaluation performance metrics

Three indicators are used to evaluate the performance metric for the developed BERT-based text classification approach in this research: (1) accuracy; (2) precision; and (3) recall, which are determined by:

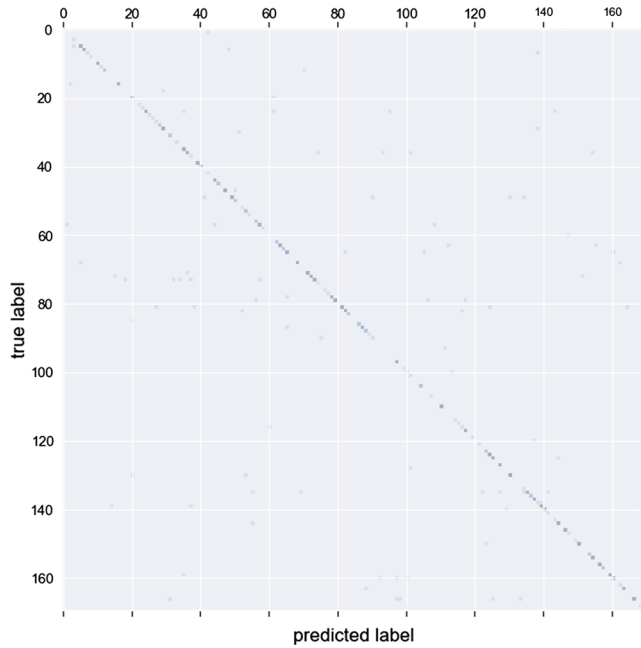
$$\text{Accuracy} = \frac{\text{Number of correctly classified classes}}{\text{Total number of classes}} \times 100\% \quad (6)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (7)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (8)$$

**Table 3**  
Examples of error classification.

| Ground-truth |                                 |                                                                                                 | Prediction |
|--------------|---------------------------------|-------------------------------------------------------------------------------------------------|------------|
| Label        | Corresponding near miss reports | Description of near miss reports                                                                | Label      |
| 144          | External electric protection    | The primary distribution cabinet guardrail is not locked                                        | 55         |
| 57           | Adjacent edge protection        | The protection measures of elevator shaft facade were removed                                   | 1          |
| 24           | Scaffold protection             | The protective barrier of the working layer does not meet the requirements of the specification | 35         |
| 135          | Foundation pit support          | Foundation pit support and protection is not timely                                             | 122        |
| 82           | Scaffold stability              | The frame is not fastened, or the rod joints are not fastened                                   | 52         |



**Fig. 4.** Confusion Matrix of classification results.

where ‘accuracy’ refers to the ratio of correctly classified near miss events in portion to their total. Similarly, the ‘precision’ means the ratio of correctly classified positive near miss event in portion to their total. Likewise, for the ‘Recall’, it is the ratio of correctly classified positive near miss event in portion to their total. ‘True positive’ (TP) refers to a near-miss event where the proposed approach makes an accurate classification. A ‘false positive (FP)’ occurs when the proposed approach classifies near miss event A (positive sample) as event B. However, a ‘false negative (FN)’ may arise when false sample is classified as positive sample.

### 5.3. Results

The parameters for training the model are set as follows: Batch size is 32, Gradient descent method is Adam, the learning rate is  $2e-5$ , and the maximum epoch for training is 100. Additional details regard with regard to the training and parameters can be found in Devlin et al. [53].

The BERT produces an output in the form of a probability vector; each instance of a near-miss report consists of a 170-dimensional vector. As our testing database contains 657 near miss reports, and the output of the model is a  $657 \times 170$ -dimensional vector. Examples of error classification have been presented in Table 3. Fig. 4 presents the confusion matrix for the classification results. From the confusion matrix, we can determine the results of the error and correct classification results. Fig. 5 presents the precision and recall for each category. In our research, the average accuracy for the near-miss text classification is

86.91%, and the average precision and recall are 51% and 54%, respectively. Notably, some near-miss reports are not correctly classified. For example, the ground-truth label of a near-miss event is 144. However, our proposed approach classifies this near-miss report to label 55. We suggest that the similarity of words in the text affects the classification performance. Also, the sample of each category is too small, leading to a lower precision value and recall value.

Five types of deep learning-based text classification approaches are compared and used to evaluate our classification performance:

1. FastText [19],
2. TextCNN and BiGRU [23,21],
3. Text Convolutional Neural Network (TextCNN) [20],
4. BiGRU and attention [34]; and
5. Text Recurrent Convolutional Neural Network (TextRCNN) [23,21].

Our comparative evaluation used the same training and testing database as noted in Section 3. In Table 4, it can be seen that our proposed deep learning approach is more accurate than other automatic text classification methods.

## 6. Discussion

We have developed a novel approach for classifying text contained with safety reports. To our knowledge, there has been a paucity of studies that have utilised deep learning to classify safety incidents, particularly near-misses. The results of the classification process can provide managers with insights into the context and nuances of near-misses. Being able to acquire context knowledge of near-misses provides managers with the ability to respond quickly to situations that may result in adverse safety outcomes. The information gleaned from the analysis of the text can be shared with people working on-site and used to educate them about ‘what not to do’ and ‘what to do’ to keep themselves on-site. Also, managers can acquire knowledge of the typical activities where near-misses are likely to occur and thus put resources and mechanism in place to improve safety levels. In sum, the classification of text from safety reports can be used to supplement a digital strategy that is being used to support the adoption of error management culture during construction [22].

Our contributions to knowledge are threefold. First, there are no available databases of near-misses that have been compiled and been subjected to text classification. Objectively comparing and evaluating the performance of our approach against other deep learning-based text classification approaches has enabled us provide a new avenue to develop and extend this field of inquiry. Secondly, we integrate the advanced NLP and deep learning approaches to classify near-miss reports by learning from a pre-trained deep bi-directional representation on both left and right contexts in all layers. In addition, we modified the BERT’s architecture (i.e. fully connected layers) and adjusted its hyperparameters. The experimental results demonstrated that our developed deep learning approach could achieve a comparable performance of text classification.

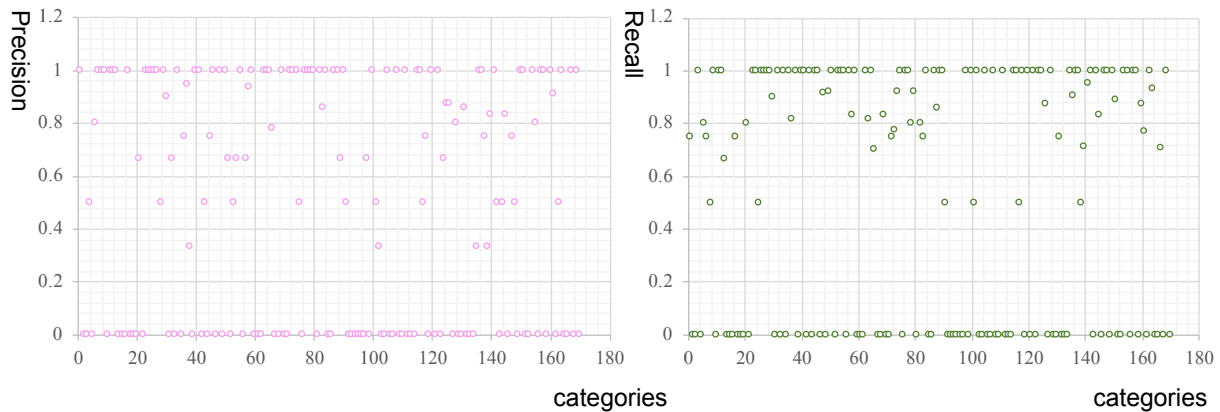


Fig. 5. Precision and Recall of the classification results.

Table 4

Comparison of our developed deep learning method and other text classification approaches.

| Algorithm | Fast Text [19] | TextCNN + BiGRU [23,21] | TextCNN [20] | BiGRU + Attention [34] | TextRCNN [23,21] | Our Developed Approach |
|-----------|----------------|-------------------------|--------------|------------------------|------------------|------------------------|
| Accuracy  | 74.63%         | 75.31%                  | 74.82%       | 75.18%                 | 75.26%           | 86.91%                 |

### 6.1. Limitations

Despite the success of our developed model to classify on near misses, the following limitations need to be acknowledged. Firstly, near-miss data contained in the safety reports were recorded and uploaded by safety personnel from different companies. People's perceptions for categorising near-misses can vary. As a result, the same near-miss type may have been classified differently. Secondly, our BERT approach applied an unsupervised method to pre-training our network. The pre-training 3.3-billion-word corpus data derived from BooksCorpus and English Wikipedia was unlabeled. Therefore, the process of pre-training the model takes a long time, especially when new categories of near-misses are identified. Thirdly, the developed model was unable to 100% accurately classify near-miss reports due to sheer number of categories ( $L = 170$ ), which contained too few events. Finally, the data source is in Chinese. In this experiment, we translated the data into English. Thus, the quality of the translation may have affected the experimental results.

## 7. Conclusion

We have developed a deep learning-based text classification approach to classify near-miss data contained within safety reports automatically. The developed BERT approach was designed in the form of bi-directional pre-training for language representation, eliminates the need for heavily-engineered task-specific architectures. The experimental results we have produced demonstrate that our approach can accurately and automatically classify near-miss data contained within safety reports.

The average accuracy of our deep learning model is 86.9%, which outperformed other state-of-the-art approaches. While we were not able to achieve 100% accuracy, the developed BERT approach can provide site management with several benefits for safety. For example, recorded near misses can be automatically categorised and stored to enable site engineers and managers to understand better the nuances of hazards that can prevail on construction sites. Further research is required to improve the accuracy of classifying safety data, particularly in the context of annotating training text. Also, future research needs to focus on creating larger datasets and using unsupervised learning to improve the accuracy of text classification.

### Declaration of Competing Interest

The authors declared that they have no conflicts of interest to this work. We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

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